

N-Channel Switch

J111, J112, J113, MMBFJ111, MMBFJ112, MMBFJ113

Features

- This Device is Designed for Low Level Analog Switching, Sample and Hold Circuits and Chopper Stabilized Amplifiers
- Sourced from Process 51
- Source & Drain are Interchangeable
- These are Pb-Free Devices

ABSOLUTE MAXIMUM RATINGS ($T_A = 25^\circ\text{C}$ unless otherwise noted) (Note 1, 2)

Symbol	Parameter	Value	Unit
V_{DG}	Drain-Source Voltage	35	V
V_{GS}	Gate-Source Voltage	-35	V
I_{GF}	Forward Gate Current	50	mA
T_J, T_{STG}	Operating and Storage Junction Temperature Range	-55 to 150	$^\circ\text{C}$

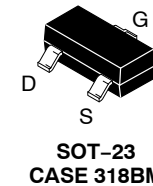
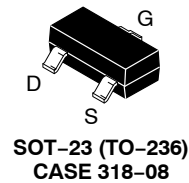
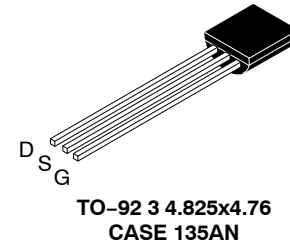
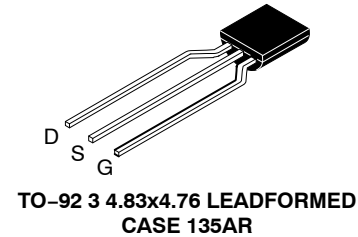
Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

1. These ratings are based on a maximum junction temperature of 150°C .
2. These are steady-state limits. ON Semiconductor should be consulted on applications involving pulsed or low-duty-cycle operations.

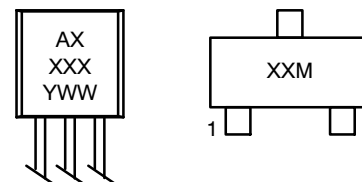
THERMAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$ unless otherwise noted)

Symbol	Parameter	Max		Unit
		J111 / J112 / J113 (Note 3)	MMBFJ111 / MMBFJ112 / MMBFJ113 (Note 4)	
P_D	Total Device Dissipation	625	350	mW
	Derate Above 25°C	5.0	2.8	mW/ $^\circ\text{C}$
$R_{\theta JC}$	Thermal Resistance, Junction-to-Case	125	-	$^\circ\text{C}/\text{W}$
$R_{\theta JA}$	Thermal Resistance, Junction-to-Ambient	200	357	$^\circ\text{C}/\text{W}$

3. PCB size: FR-4, 76 mm x 114 mm x 1.57 mm (3.0 inch x 4.5 inch x 0.062 inch) with minimum land pattern size.
4. Device mounted on FR-4 PCB 36 mm x 18 mm x 1.5 mm; mounting pad for the collector lead minimum 6 cm^2 .



MARKING DIAGRAMS



XXXX, XX = Specific Device Code
A = Assembly Plant Code
Y = Year
WW = Work Week
M = Date Code

ORDERING INFORMATION

See detailed ordering and shipping information on page 6 of this data sheet.

J111, J112, J113, MMBFJ111, MMBFJ112, MMBFJ113

ELECTRICAL CHARACTERISTICS (T_J = 25°C unless otherwise noted)

Symbol	Parameter	Test Condition	Min	Max	Unit
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OFF CHARACTERISTICS

V _{(BR)GSS}	Gate–Source Breakdown Voltage	I _G = –1.0 μA, V _{DS} = 0		–35	–	V
I _{GSS}	Gate Reverse Current	V _{GS} = –15 V, V _{DS} = 0		–	–1.0	nA
V _{GS(off)}	Gate–Source Cut–Off Voltage	V _{DS} = 5 V, I _D = 1.0 μA	111	–3.0	–10.0	V
			112	–1.0	–5.0	
			113	–0.5	–3.0	
I _{D(off)}	Drain Cutoff Leakage Current	V _{DS} = 5.0 V, V _{GS} = –10 V		–	1.0	nA

ON CHARACTERISTICS

I _{DSS}	Zero–Gate Voltage Drain Current (Note 5)	V _{DS} = 15 V, V _{GS} = 0	111	20	–	mA
			112	5.0	–	
			113	2.0	–	
r _{DS(on)}	Drain–Source On Resistance	V _{DS} ≤ 0.1 V, V _{GS} = 0	111	–	30	Ω
			112	–	50	
			113	–	100	

SMALL SIGNAL CHARACTERISTICS

C _{dg(on)} C _{sg(on)}	Drain–Gate & Source–Gate On Capacitance	V _{DS} = 0, V _{GS} = 0, f = 1.0 MHz	–	28	pF
C _{dg(off)}	Drain–Gate Off Capacitance	V _{DS} = 0, V _{GS} = –10 V, f = 1.0 MHz	–	5.0	pF
C _{sg(off)}	Source–Gate Off Capacitance	V _{DS} = 0, V _{GS} = –10 V, f = 1.0 MHz	–	5.0	pF

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

5. Pulse test: pulse width ≤300 μs, duty cycle ≤2%.

TYPICAL PERFORMANCE CHARACTERISTICS

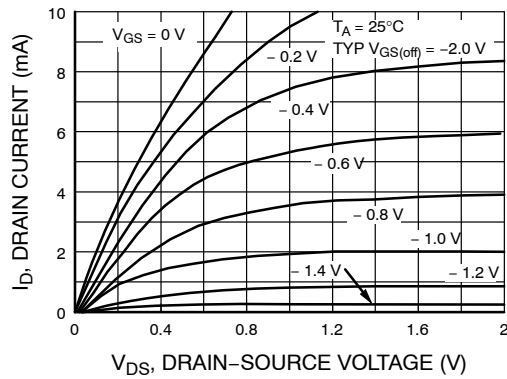


Figure 1. Common Drain-Source

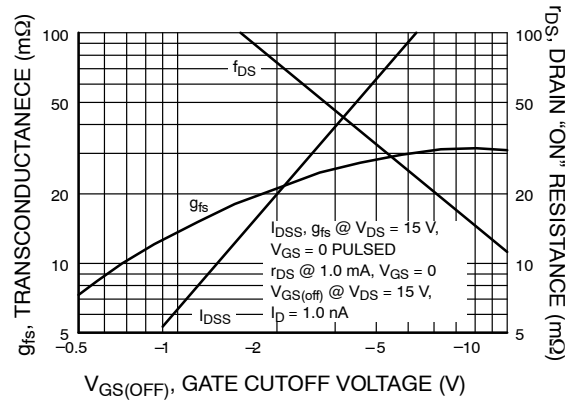


Figure 2. Parameter Interactions

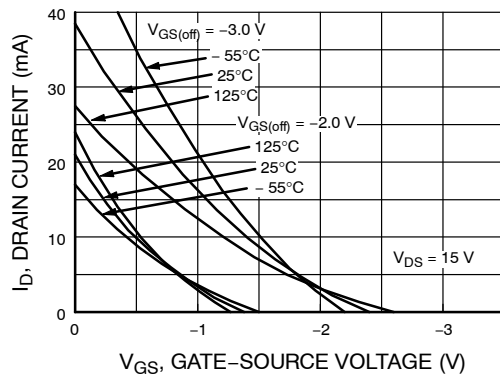


Figure 3. Transfer Characteristics

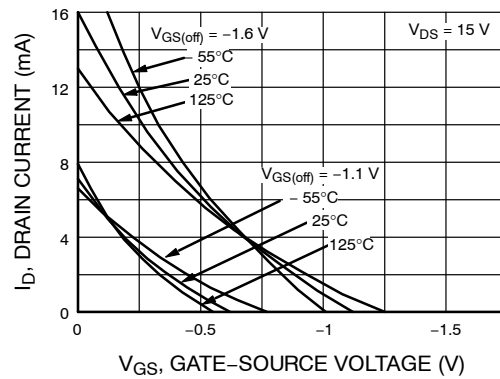


Figure 4. Transfer Characteristics

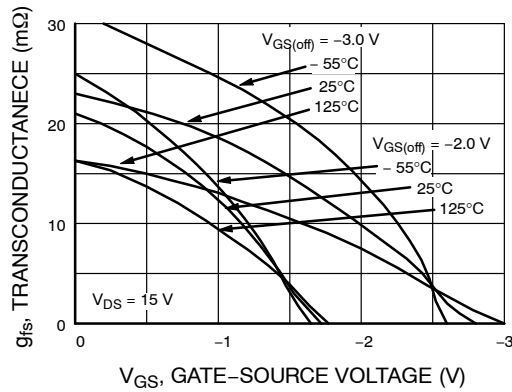


Figure 5. Transfer Characteristics

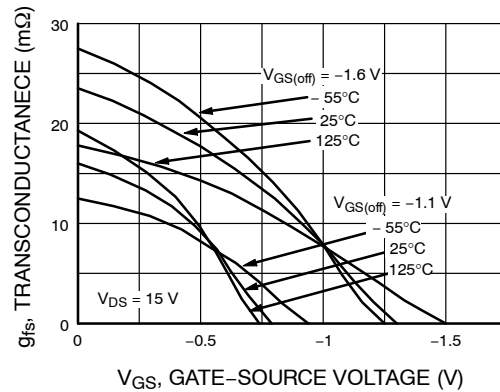


Figure 6. Transfer Characteristics

TYPICAL PERFORMANCE CHARACTERISTICS (CONTINUED)

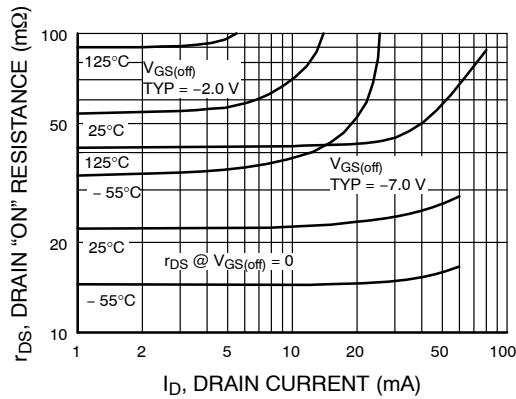


Figure 7. On Resistance vs. Drain Current

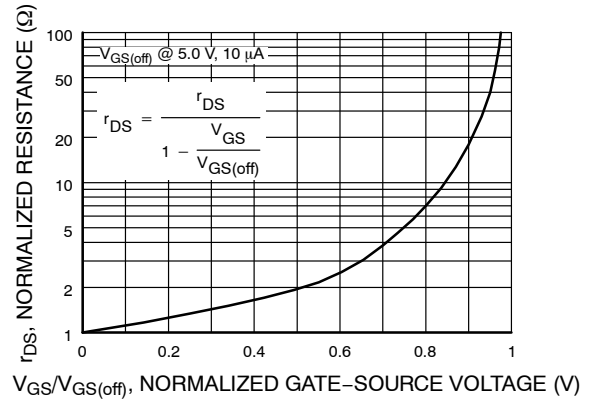


Figure 8. Normalized Drain Resistance vs. Bias Voltage

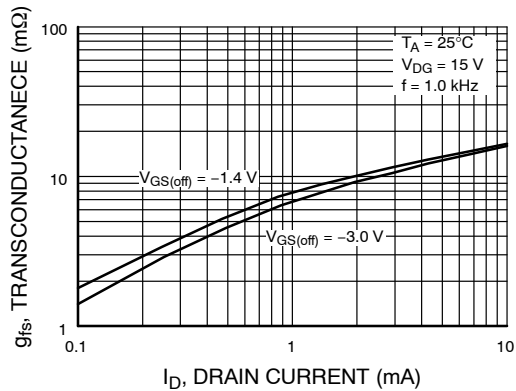


Figure 9. Transconductance vs. Drain Current

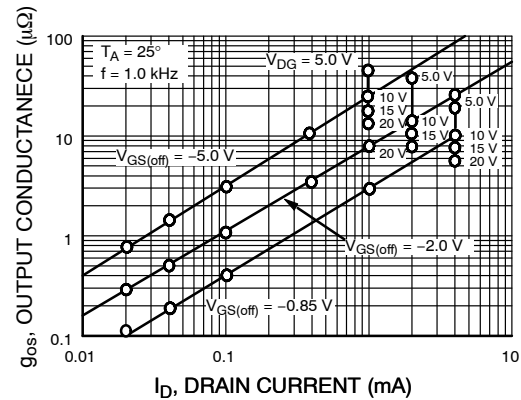


Figure 10. Output Conductance vs. Drain Current

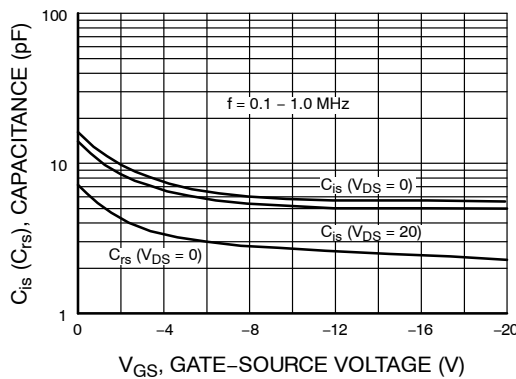


Figure 11. Capacitance vs. Voltage

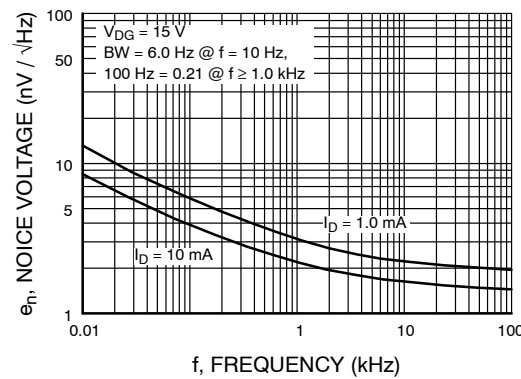


Figure 12. Noise Voltage vs. Frequency

TYPICAL PERFORMANCE CHARACTERISTICS (CONTINUED)

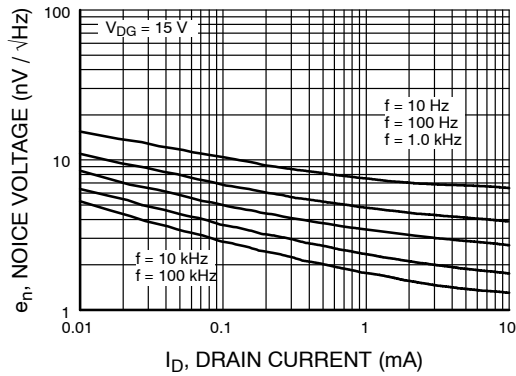


Figure 13. Noise Voltage vs. Current

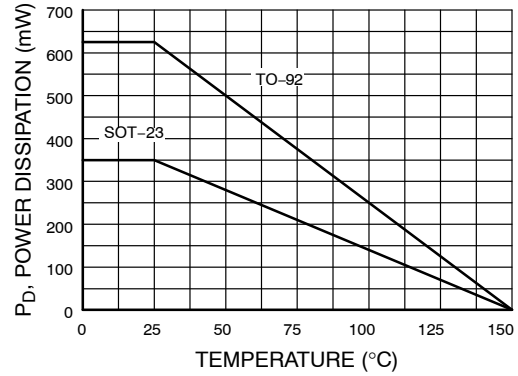


Figure 14. Power Dissipation vs. Ambient Temperature

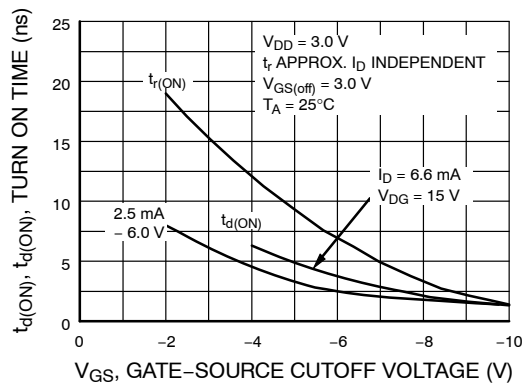


Figure 15. Switching Turn-On Time vs. Gate-Source Voltage

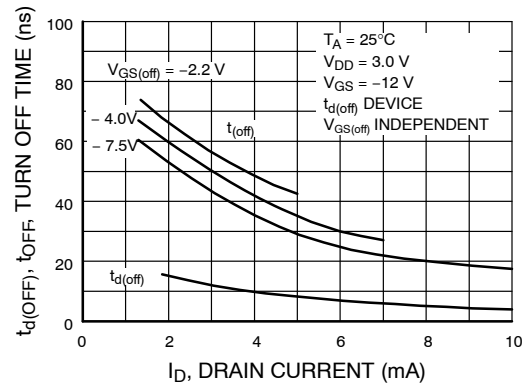


Figure 16. Switching Turn-Off Time vs. Drain Current

J111, J112, J113, MMBFJ111, MMBFJ112, MMBFJ113

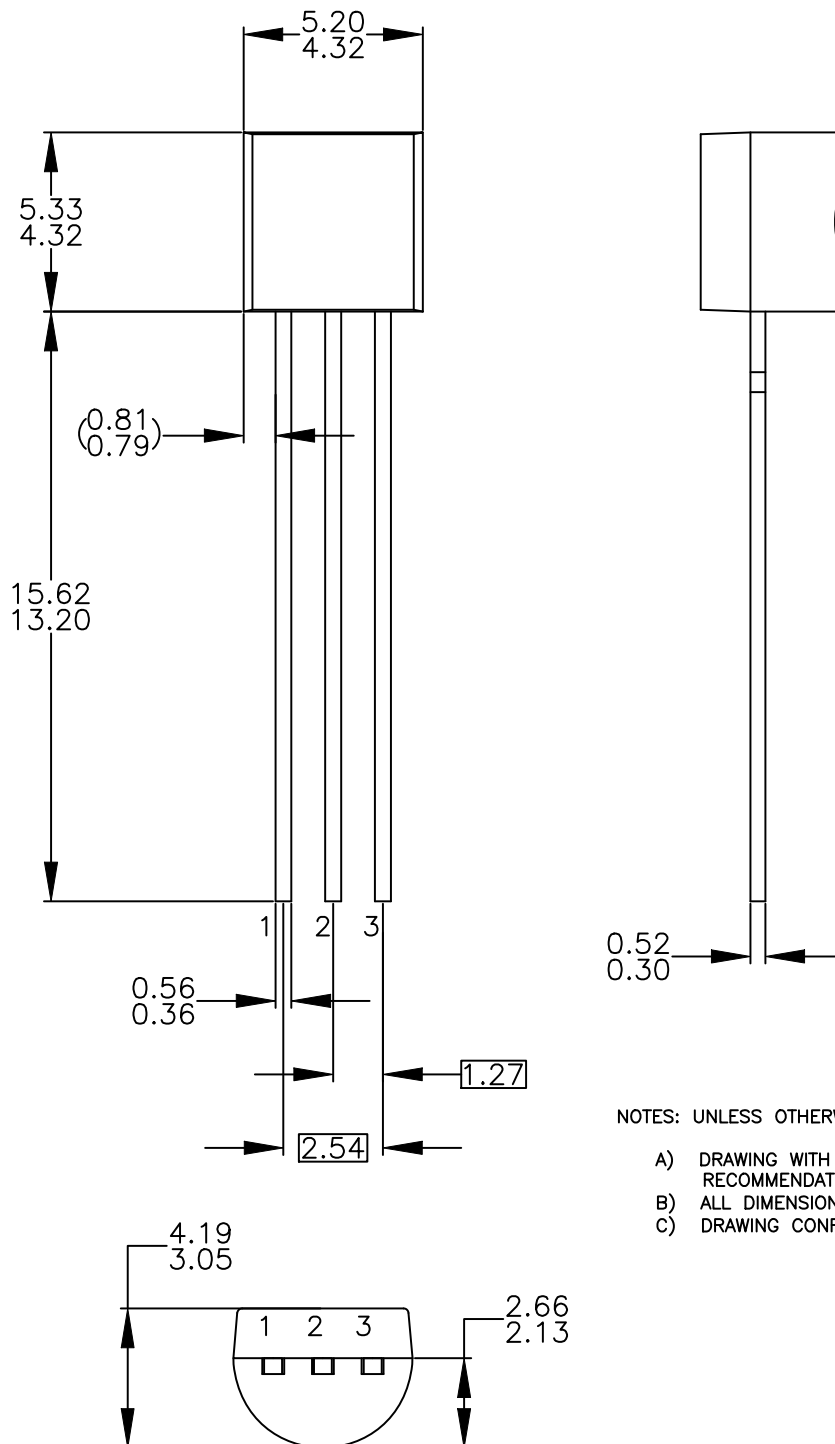
ORDERING INFORMATION

Part Number	Top Mark	Package	Shipping [†]
J111	AJ 111 YWW	TO-92 3L (Pb-Free)	10000 Units / Bulk
J111-D26Z	AJ 111 YWW	TO-92 3L (Pb-Free)	2000 / Tape & Reel
J111-D74Z	AJ 111 YWW	TO-92 3L (Pb-Free)	2000 / Ammo
J112	AJ 112 YWW	TO-92 3L (Pb-Free)	10000 Units / Bulk
J112-D26Z	AJ 112 YWW	TO-92 3L (Pb-Free)	2000 / Tape & Reel
J112-D27Z	AJ 112 YWW	TO-92 3L (Pb-Free)	2000 / Tape & Reel
J112-D74Z	AJ 112 YWW	TO-92 3L (Pb-Free)	2000 / Ammo
J113	AJ 113 YWW	TO-92 3L (Pb-Free)	10000 Units / Bulk
J113-D74Z	AJ 113 YWW	TO-92 3L (Pb-Free)	2000 / Ammo
MMBFJ111	6P	SOT-23 3L (Pb-Free)	3000 / Tape & Reel
MMBFJ112	6R	SOT-23 3L (Pb-Free)	3000 / Tape & Reel
MMBFJ113	6S	SOT-23 3L (Pb-Free)	3000 / Tape & Reel

[†]For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

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DATE 31 JUL 2016



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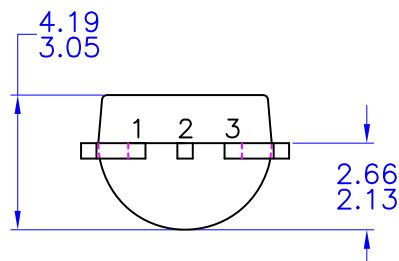
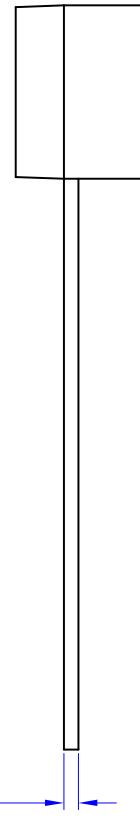
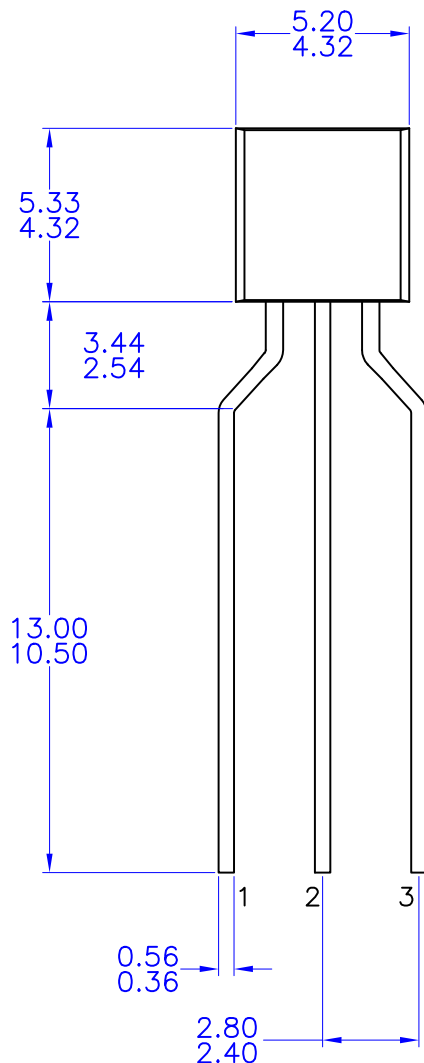
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